

Minkyu Jun

mkjun@umich.edu | (651) 359-3599 | www.linkedin.com/in/minkyu-jun

EDUCATION

University of Michigan — Ann Arbor

Master of Science in Engineering, Computer Science

Bachelor of Science in Engineering, Computer Science (**GPA: 4.00**)

Ann Arbor, MI

Expected Dec 2026

Expected Dec 2025

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Organization, Database Management System, Web Systems, Software Engineering, Accessible Computing

SKILLS

- **Languages:** C++, Python, Java, JavaScript, TypeScript, HTML5, CSS, SQL
- **Technologies and Frameworks:** React, React Native, Flask, Django, PostgreSQL, MongoDB, Firestore, AWS
- **Developer Tools:** Git, Github, Linux, Visual Studio Code, IntelliJ, Jira, Figma

WORK EXPERIENCE

Epic Systems

Software Developer Intern

Verona, WI

May 2025 — Aug 2025

- Developed a custom **AI agent** with **OpenAI GPT-4.1** model and **prompt engineering** to convert users' natural language instructions into structured JSON, automating the generation of scheduling rules without manual configuration and reducing setup time by **75%** while achieving **90%** accuracy.
- Architected end-to-end workflow and engineered **AI support chatbot** systems that dynamically route user queries to specialized AI agents, minimizing the time users spend interpreting and refining AI-generated configurations.
- Designed and created a **responsive UI in React and TypeScript** that dynamically maps AI-generated JSON to relevant interface sections, providing clear, real-time feedback on AI-driven updates.

University of Michigan Center for Healthcare Engineering & Patient Safety (CHEPS)

Software Engineering Intern

Ann Arbor, MI

Aug 2024 — Jan 2025

- Implemented linear programming with **C++** to build a scheduling optimization tool that prioritized a hierarchy of metrics, reducing manual scheduling time for **250+** Michigan Medicine clinicians by **70%**.
- Migrated from IBM CPLEX solver to the in-house OptiCore library, improving maintainability across modules.
- Designed unit tests and led debugging sessions to verify schedule feasibility and minimize edge-case failures.
- Adopt Agile methodology with **Jira**, leading biweekly sprints for efficient task tracking and team collaboration.

PROJECT EXPERIENCE

Open Course | React, JavaScript, Flask, Python, PostgreSQL

- Built a server-side CRUD REST API using **Flask** that allowed users to create profiles, post group study plans (including course name, meeting time, and location), and join existing groups.
- Crafted post and filter options pages using **React** and **Tailwind CSS** to enhance user experience by allowing users to view study group posts and search for groups based on specific criteria.
- Collaborated with the design team to identify and address UI/UX concerns, aligning the development with high-fidelity Figma wireframes.

Course Flowchart | React, JavaScript, Coursedog API, Mermaid.js

- Devised customized pattern-recognition algorithms with **JavaScript** to parse and establish “and/or” logic for prerequisite information directly from Coursedog Curriculum API's course descriptions.
- Applied **Depth-First Search (DFS)** algorithms and Mermaid.js to display graph that visually illustrates the flow of major's curriculum and prerequisite information for each class.

TEACHING EXPERIENCE

University of Michigan Department of Computer Science & Engineering

Database Management System (EECS 484) Teaching Assistant

Ann Arbor, MI

Aug 2025 — Present

- Assisted in teaching an upper-level computer science course with **300+** students, covering ER diagrams, **SQL**, **Java**, **MongoDB**, query optimization, advanced sorting techniques, and crash recovery.
- Led weekly discussion sessions for **30+** students, assisted in developing curriculum and coursework materials, and held office hours to support students with homework, projects, and exam preparation.